Credit Statement : We consulted the lecture notes and did not work with anyone outside of our group.

Conclusion:

Based on each algorithms Big O speed merge is the fastest with a worst case of O(n logn) while both Lomuto's partition and Hoare's partition are O(n^2). This makes us think that the fastest algorithm would be Merge Sort in average case, depending on the structure of the array to sort. From the experiment, the fastest algorithm was Hoare, followed closely by Lomuto, and finally Merge Sort. This is likely due to the fact that the memory usage of the two algorithms with QS is O(log n) which should be the way that the clock function works, therefore showing a faster computation. The reason Hoare was fastest was likely due to the way that the array was organized, which allowed it to sort slightly faster than Hoare.